

CCNA 200-301 Notes - Chapter 12: VLSM

Definition

VLSM (Variable Length Subnet Mask) allows different subnet masks inside the same network to minimize IP address waste.

Why Use VLSM?

- Reduce IP address waste.
- Allocate only the required number of hosts.
- More efficient than Fixed Length Subnetting.
- Commonly used in enterprise networks.

Golden Rule

Always allocate the largest subnet first, then continue with smaller subnets.

Example

Original Network: 192.168.1.0/24

Hosts	CIDR	Mask	Usable Hosts
60	/26	255.255.255.192	62
30	/27	255.255.255.224	30
14	/28	255.255.255.240	14
6	/29	255.255.255.248	6
2 (Router Link)	/30	255.255.255.252	2

Steps

1. Sort requirements from largest to smallest.
2. Choose the proper subnet mask.
3. Assign the subnet.
4. Continue from the next available network.
5. Repeat until finished.

Quick Notes

- VLSM is based on subnetting.
- Use CIDR notation.
- Every subnet starts after the previous broadcast address.
- Different subnet masks are allowed.